CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 9: Some Applications of Trigonometry

Previous year board questions, topic-wise: 16 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 One right-angled triangle

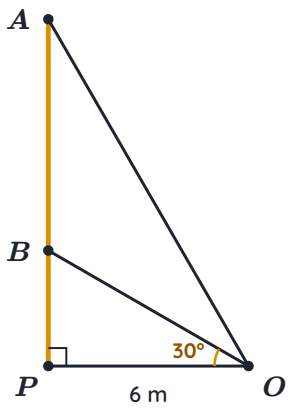
1. A car drives away from the foot of a tower 30 m high. What is the angle of elevation of the top of the tower from the car when it is $10\sqrt{3}$ m from the foot? (a) 30° (b) 45° (c) 90° (d) 60° CBSE 2026, 1 mark

2. A kite is flying 150 m above the ground. Its string is straight and makes an angle of 30° with the ground. How long is the string? CBSE 2025, 1 mark

3. A ladder leaning against a wall makes an angle of 60° with the ground. Its foot is 2.5 m from the wall. Find the length of the ladder. CBSE 2016, 1 mark

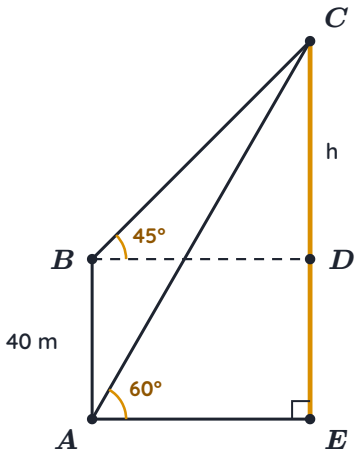
2 Two right-angled triangles

4. A radio tower PA is built in two sections: PB is the lower section and BA the upper one. Wires from a point O on the ground, 6 m from the foot P, hold the tower at B and at A. From O the angle of elevation of B is 30° and that of A is 60°. (i) Find the length of the wire OB. (ii) Find the length of the wire OA. (iii) Find AB, OR find the area of $\triangle OPB$. CBSE 2026, 4 marks



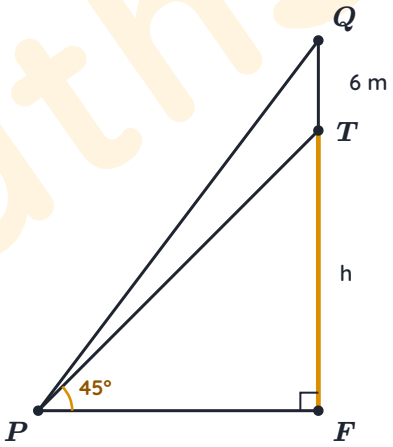
Angle POB = 30° and angle POA = 60°

5. From a point A on the ground, the top C of a lighthouse CE is seen at an angle of elevation of 60°. From a deck at B, 40 m directly above A, the angle of elevation of C is 45°. D is the point on CE level with B, and $CD = h$ metres. (i) Write BD in terms of h. (ii) Write BC in terms of h. (iii) Find the height CE, taking $\sqrt{3} = 1.73$, OR find AE if $AC = 100$ m. CBSE 2025, 4 marks



6. A 6 m pole stands on top of a tower. From a point P on the ground, the top of the pole is seen at an angle of elevation of 60°. From the top of the tower, P is seen at an angle of depression of 45°. Find the height of the tower and the distance of P from its foot. (Take $\sqrt{3} = 1.73$.)

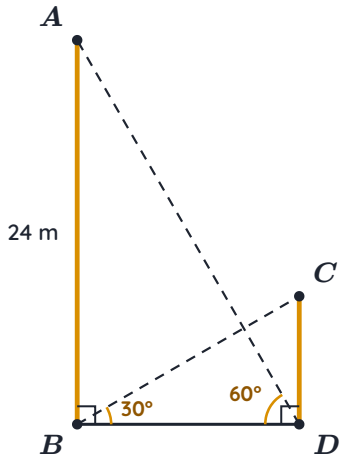
CBSE 2024, 5 marks



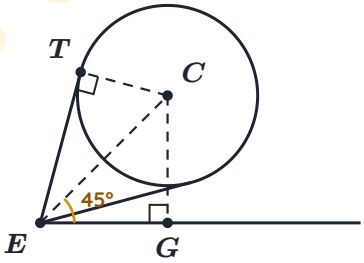
FT is the tower and TQ is the pole

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7. Two towers stand on level ground. The first is 24 m high, and from the foot of the second tower its top is seen at an angle of elevation of $60°$. From the foot of the first tower, the top of the second is seen at an angle of elevation of $30°$. Find the distance between the towers, the height of the second tower, and the length of a wire joining the two tops. CBSE 2023, 5 marks

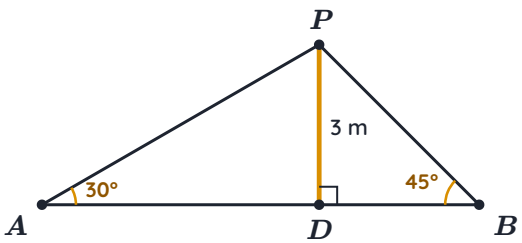


8. A spherical balloon of radius r subtends an angle of $60°$ at the eye of an observer, and the angle of elevation of its centre from the eye is $45°$. Prove that the centre of the balloon is at a height of $\sqrt{2}\,r$. CBSE 2023, 5 marks

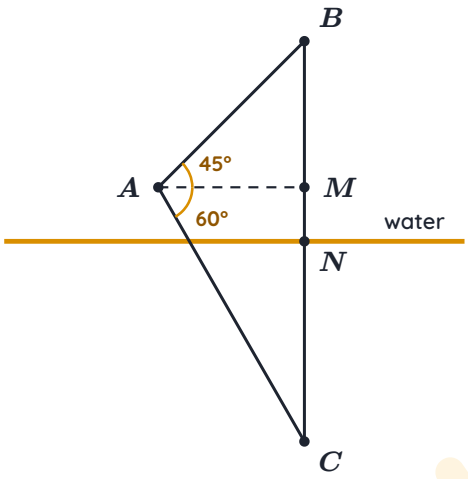


9. A tower is 50 m high. From the foot of a building, the top of the tower is seen at an angle of elevation of $60°$. From the foot of the tower, the top of the building is seen at an angle of elevation of $30°$. Find the height of the building. CBSE 2022 Term 2, 3 marks

10. From a point on a bridge 3 m above the banks of a river, the two banks on opposite sides are seen at angles of depression of $30°$ and $45°$. Find the width of the river. CBSE 2022 Term 2, 3 marks

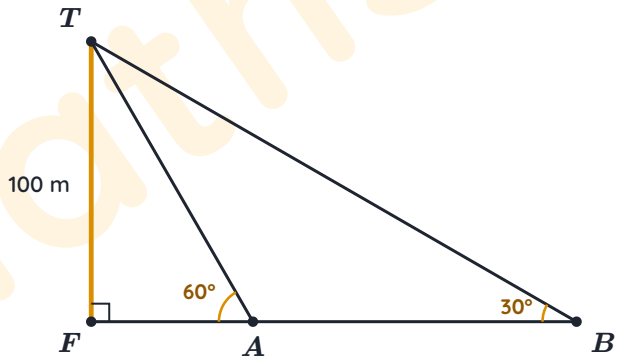


11. The top B of a pavilion standing in a lake is 10 m above the water. From a point A, h metres above the water, the angle of elevation of B is 45° and the angle of depression of the reflection C of B in the water is 60°. (i) Draw a labelled figure. (ii) Find h, taking $\sqrt{3} = 1.73$. CBSE 2022 Term 2, 4 marks

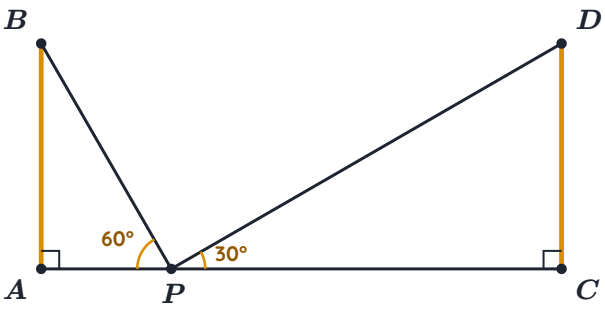


N is on the water surface, with BN = NC = 10 m

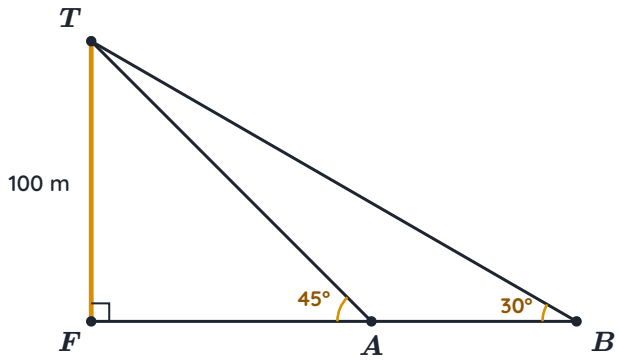
12. A man rows a boat directly away from a lighthouse 100 m high. In 2 minutes the angle of elevation of the top of the lighthouse changes from 60° to 30°. Find the speed of the boat in metres per minute. Take $\sqrt{3} = 1.732$. CBSE 2019, 4 marks



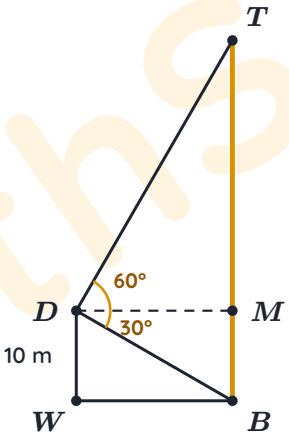
13. Two poles of equal height stand opposite each other on the two sides of a road 80 m wide. From a point on the road between them, the tops of the poles are seen at angles of elevation of 60° and 30°. Find the height of the poles and the distances of the point from them. CBSE 2019, 4 marks



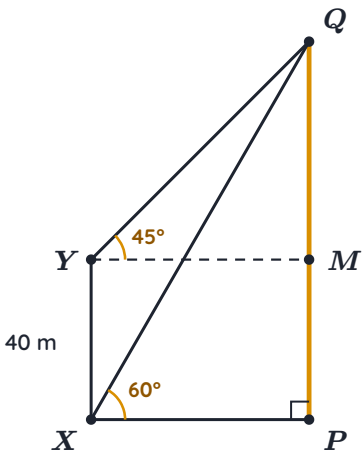
14. From the top of a lighthouse 100 m above sea level, two ships are seen at angles of depression of 30° and 45°. One ship is directly behind the other on the same side of the lighthouse. Find the distance between the ships. Take $\sqrt{3} = 1.732$. CBSE 2018, 4 marks



15. A man stands on the deck of a ship, 10 m above the water. He sees the top of a hill at an angle of elevation of 60° and the foot of the hill at an angle of depression of 30°. Find the distance of the hill from the ship and the height of the hill. CBSE 2016, 3 marks



16. The top Q of a vertical tower PQ is seen at an angle of elevation of 60° from a point X on the ground. From a point Y, 40 m vertically above X, the angle of elevation of Q is 45°. Find the height PQ and the distance PX. Take $\sqrt{3} = 1.73$. CBSE 2016, 4 marks



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